



Air movement acceptability limits and thermal comfort in Brazil's hot humid climate zone

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ABSTRACT

In hot humid climates, natural ventilation is an essential passive strategy in order to maintain thermal comfort inside buildings and it can be also used as an energy-conserving design strategy to reduce building cooling loads by removing heat stored in the buildings thermal mass. In this context, many previous studies have focused on thermal comfort and air velocity ranges. However, whether this air movement is desirable or not remains an open area. This paper aims to identify air movement acceptability levels inside naturally ventilated buildings in Brazil. Minimal air velocity values corresponding to 80% and 90% (V80 and V90) air movement acceptability inside these buildings. Field experiments were performed during hot and cool seasons when 2075 questionnaires were filled for the subjects while simultaneous microclimatic observations were made with laboratory precision. Main results indicated that the minimal air velocity required were at least 0.4 m/s for 26 °C reaching 0.9 m/s for operative temperatures up to 30 °C. Subjects are not only preferring more air speed but also demanding air velocities closer or higher than 0.8 m/s ASHRAE limit. This dispels the notion of draft in hot humid climates and reinforce the broader theory of alliesthesia and the physiological role of pleasure due to air movement increment.

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1. Introduction

Human perception of air movement depends on air velocity, air velocity fluctuations, air temperature, and personal factors such as overall thermal sensation, clothing insulation and physical activity level (metabolic rate) [1]. Air velocity affects both convective and evaporative heat losses from the human body, and thus influences thermal comfort conditions [2].

If we agree that thermal environments that are slightly warmer than preferred or neutral can still be acceptable to building occupants, as the adaptive comfort model suggests [3], then the introduction of airflow with higher velocities into such environments might be universally regarded as desirable. Higher velocities' effect will be to remove sensible and latent heat from the body, so body temperatures will be restored to their comfort set-points. This

hypothesis can be deduced from the physiological principle of *alliesthesia* [4].

Alliesthesia describes the phenomenon whereby a given stimulus can induce either a pleasant or unpleasant sensation, depending on the subject's internal state [4]. The observation that cold receptors are closer to the skin surface than warm receptors explains why draft represents an *unpleasant* stimulus (negative alliesthesia) in cold environments whereas the same level of air movement is perceived as pleasant (positive alliesthesia) in warm environments. It also renders illogical the notion of draft in warm environments, which accounts for the widely reported inadequacy of the Fanger et al. [5], Draft Risk (DR) at explaining air movement preferences of occupants' into warm environments [6].

Many previous studies have attempted to define when and where air movement is either desirable or not desirable [7–11]. Thermal comfort research literature indicates that indoor air speed in hot climates should be set between 0.2 and 1.50 m/s, yet 0.2 m/s has been deemed in ASHRAE Standard 55 [12] to be the threshold of draft perception inside air-conditioned buildings where occupants have no direct control over their environment [12]. None of the previous research explicitly addressed air movement acceptability,

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instead focusing mostly on overall thermal sensation and comfort [1].

Based on their experiments with occupant controlled air movement in climate chamber, Fountain et al. [13] suggested an index, the PS model. PS model is a model of “predicted percent of satisfied people” as a function of locally controlled air movement in the occupied zone [13]. Providing the “percent satisfied” at a specific operative temperature and air velocity, this model offered a different approach focusing on the preferred air velocity rather than limits as the draft risk suggested. However, subjects’ conditions in climate chambers differ into real buildings and therefore experiments into indoor environments could provide complementary results [14].

Much of Brazil’s territory is classified as having a hot humid climate. In such regions, natural ventilation combined with solar protection are the most effective building design strategies to achieve thermal comfort without resorting to mechanical cooling. Despite these favourable conditions, the number of buildings using air-conditioned systems as main cooling design strategy has been dramatically increasing. Based on this scenario and the recent energy crises [15] Brazilian Government has been promoting energy conservation initiatives including a recent Federal Regulation for Voluntary Labelling of Energy Efficiency Levels in Commercial, Public and Service Buildings [16]. This new regulation summarizes an immense effort in order to provide guidelines based on Brazil’s climate requirements for designers in general with specific items related to lighting system, HVAC and building envelope. However, naturally ventilated indoor environments still appears as an open category and the references into this proposed regulation refers direct to current standards such as ASHRAE [12].

This paper is focused on the relationship between air movement acceptability and thermal comfort, inside naturally ventilated buildings in the north-east of Brazil (Maceio city). This research aims to define minimum air speeds necessary to produce 80% and 90% acceptability levels for the occupants of naturally ventilated buildings in hot humid climates.

2. Method

The method adopted for this work is based on analysis of relationship between air movement acceptability and thermal comfort inside naturally ventilated buildings located in the north-east of Brazil (Maceio city). This method is based on design proximal indoor climate data with simultaneous questionnaires filled in by occupants of naturally ventilated spaces. A survey including 2075 questionnaires¹ during the cool (August–September) and also hot season (February–March) was carried out, where subjects were asked to inform their thermal preferences while microclimatic measurements were taken.

2.1. Maceio’s climatic environment

Maceio is located on the north-east sea coast of Brazil (latitude 9°40’ south of Equator and longitude 35°42’ west of Greenwich). The low latitude combined with high solar radiation intensity, as well as the proximity of large warm water surfaces – ocean and lagoons – elevates the humidity level, hence the climate is classified as hot and humid, Aw, according to Köppen’s classification.

Maceio’s climate is very equable, which is typical of the north-east coast of Brazil. Seasons are divided into just two: winter and summer, although the “winter” remains warmer than many mid-latitude climate zones’ summers. Because of this, summer can be classified as a *hot season* and the winter as a *cool season* (these descriptions will be applied for this paper). The mean annual temperature is around 26 °C and the annual thermal amplitude is 3.4 °C (the highest monthly average occurs in February – 26.7 °C and the lowest monthly average is in July – 23.7 °C). Typically, the hottest days occur from November to February and the coolest days from June to August.

The mean relative humidity is around 78% during hot season and 84% during cool season. However, it is possible to encounter saturation (100%) during cooler, rainier periods. The annual average rainfall is around 1654 mm and the typical rainy season occurs from April to July. Maceio is under the influence of the south-east and north-east trade winds within the planets broadscale general atmospheric circulation.

2.2. Measurement rooms and subjects’ profile

When choosing the indoor environments for this study the following criteria were used: (i) windows had to be easy to access and operate; (ii) rooms could not have a mechanical cooling system (refrigerated air-conditioning); (iii) rooms could have complementary mechanical ventilation with unconditioned air (fans); (iv) opening and closing of windows had to be the primary means of regulating thermal conditions, and the occupants had to be engaged in near sedentary activity (1–1.3 met), and had to be permitted to freely adapt their clothing to the indoor and/or outdoor thermal conditions. Some rooms of the Federal University of Alagoas and the Centre of Superior Studies of Alagoas fitted these selection criteria and were chosen for this survey.

Monitored rooms were classrooms that were also used for drawing activities (studios) and normally occupied about 20 students; see Fig. 1 and Appendix A. In addition, the buildings presented large open spaces and natural ventilation was intentionally the main cooling strategy. In both buildings, windows were easily controlled collectively by the occupants and ceiling fans provided supplemental air movement inside the rooms.

Subjects were, on average, 21 years old, weighed 59 kg and 1.7 m in height. The sample of respondents reflected the gender imbalance of Brazil’s architecture student population, and was biased towards females. Table 1 summarizes subjects’ profile details for each season. Activities performed by the occupants of these environments were assessed as sedentary with a variation between 58 and 93 W/m² because the subjects usually stayed seated whilst drawing or writing, see Fig. 1. The clothes were light – around 0.30 clo during the hot season and 0.50 clo during the cool season (see Fig. 2a and b), as estimated from clothing garment checklists in ASHRAE Standard 55 [12].

2.3. Measurement equipment

In order to measure the ambient variables this research used a microclimatic station (Babuc A), hot wire anemometer and a surface temperature thermometer. This microclimatic station is able to take measurements and store the data collected into a data logger during the measurement period. In addition, instruments such as globe thermometer, psychrometer (dry and wet-bulb temperatures) and hot wire anemometer, were also applied (see Fig. 3a)

For air speed measurements near to participants hot wire anemometers were used. The equipment was portable, and had an Airflow Developments, model TA35 sensor, see Fig. 3b. The

¹ This value corresponds to valid questionnaires after data treatment and therefore does not consider those with any sort of problem (such as incomplete answers, occupants that left the room during the measurements, etc). The total number before the data treatment was 2099 questionnaires.



Fig. 1. Classrooms (a, c) and studios (b, d) at Federal University of Alagoas (above) and Centre of Superior Studies of Alagoas (below).

Table 1
Occupants' profile per season.

	Season	Hot	Cool
	Sample size	915	1160
Gender	Female	79%	66%
	Male	21%	34%
Age (year)	Ave	21	20.8
	Min	16	17
	Max	30	30
Height (m)	Ave	1.70	1.70
	Min	1.50	1.50
	Max	2.00	1.80
Weight (kg)	Ave	59.1	59.5
	Min	42	40
	Max	99	100

minimum air speed threshold was 0.05 m/s, with resolution of 0.01 m/s. The probe registered the maximum, minimum and average values of the air speed, and also indicated the standard deviation within the 5-min sample interval. The portable hot wire anemometer was a unidirectional type, so smoke sticks were used to discern the predominant airflow before the anemometer was positioned near the subject, Fig. 3c.

2.4. Experimental procedures

The concept of this research design proximal indoor climate data with simultaneous questionnaires filled in by occupants of naturally ventilated spaces. Details related to indoor climate data and questionnaires are given below.

2.4.1. Indoor climate data

Measurements were taken during morning and afternoon lectures, for at least 2 h in each period. The subjects' activities were not interrupted in order to characterize the typical use of rooms and studios, and they were also allowed to normally use ceiling fans, task lighting and also control the openings (to close or to open doors), as described previously.

The microclimatic station was located in the centre of the room and regulated to cater for two heights. The first height was 0.60 m, corresponding to the subjects' waist height inside the classrooms. The second height was 1.10 m which corresponded to the subjects' waist height while seated in the studio classrooms. The measurements recorded were averages of 5 min. The sensors on the microclimatic station measured air and globe temperatures, air speed and humidity.

The air speed measurements close to the subjects were taken simultaneously whilst they filled out the questionnaire. For each subject, the portable hot wire anemometer was located within a 1 m radius and at the same work plan height. As a result, mean air velocities were recorded for each subject. The hot wire



Fig. 2. Occupants' typical clothes for (a) hot and (b) cool seasons.

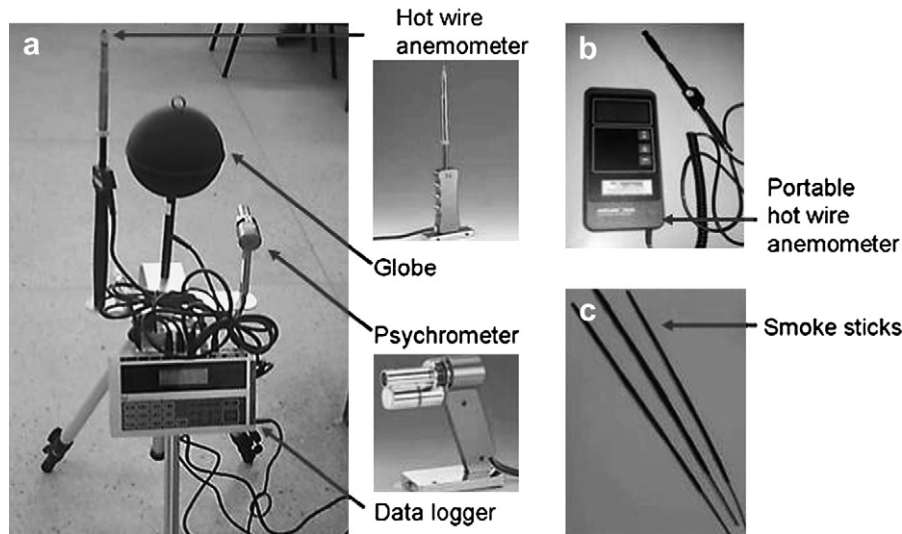


Fig. 3. (a) Microclimatic station Babuc, (b) hot wire anemometer and (c) smoke sticks.

anemometer was oriented according to the dominant flow direction indicated by the smoke sticks.

2.4.2. Questionnaire

The questionnaire aimed to characterize whole body thermal comfort and also to identify the subjects' air speed and air movement acceptability. The questionnaire was presented in three parts (see Appendix B)². The first corresponded to the subjects' demographic and anthropometric characteristics such as age, height, weight and gender. The second part included questions relating to thermal comfort, air movement acceptability and also their pattern of air-conditioning usage. In the thermal comfort part, subjects were asked about their own thermal comfort condition, their personal preferences and also about the room itself, at the time of questionnaire.

The air movement questions focused on air movement acceptability as it related to the air speed. In this case, subjects registered if the air velocity was "acceptable" or "unacceptable" and also if it was "too low" or "too high" air velocity. This specific questionnaire item is represented in Table 2.

The third and last part of the questionnaire related to the subjects' activities during the hour prior to the measurement process. It also recorded information about the subjects' clothing by way of a garment checklist. The subjects started answering the questionnaire at least half hour after they arrived in the room in order to avoid any influence of their previous activities. Each subject answered the questionnaire on five separate occasions during the same experiment.

Table 2
Air movement acceptability scale.

–2	–1	0	1	2
Unacceptable		Acceptable		Unacceptable
Because too low air velocity	But too low air velocity	Enough air velocity	But too high air velocity	Because too high air velocity

² The questionnaire version presented into this paper is a translation from Portuguese to English. In order to consider semantics' implications [21,22] the Portuguese version of this questionnaire was tested and refined during pilot surveys before the final experiments presented and discussed into this paper.

2.5. Statistical treatment

The statistical approach applied for this project followed commonly applied into this field. For air movement acceptability analysis the categorical data required a different treatment. Particularly for this data, probit analysis was conducted rather than linear regression [17]. In order to conduct these analysis, separate probit procedures were developed with software SAS[®] for each operative temperature and air velocity range. The fitted probit models achieved statistical significance at the $p = 0.05$ level and the final result is discussed into this paper.

3. Results

The percentage of subjects who indicated thermal sensations of "neutral" represented more than 60% for cool season and less than 40% for the hot season. Less than 20% of the subjects reported that they were "slightly cool" or "slightly warm" during the cool season survey (see Fig. 4). In this same season, only 3% of all subjects indicated that they were "cold" or "hot". During the hot season survey, at least 34% of the subjects indicated that were "slightly warm", and "warm" was registered more than 20% of cases. Less than 3% classified their thermal sensation as "hot". Regarding specific thermal acceptability assessments, the levels were approximately 90% for both seasons (see Fig. 5). These results met the 90% acceptability goal considered as "a higher standard of thermal comfort" [12].

Fig. 6 presents simultaneous assessments of overall thermal sensation and air movement preferences. The subjects asking for "more air movement" were trying to restore their thermal sensation towards zero (neutral). The opposite situation, when subjects preferred "less air movement" is connected with cool or cold thermal sensations. However, the number of subjects for both groups is significantly different as indicated to the "n" underneath the thermal sensation scale in Fig. 6. The majority of subjects were concentrated in thermal sensations of "slightly warm" and "warm" associated with a majority requesting "more air movement".

In an attempt to identify subjects' overall thermal dissatisfaction with their thermal environment, they were asked to indicate whether they would prefer to feel warmer or cooler. Fig. 7 summarizes thermal preference votes. Most subjects' thermal preferences were "no change" and "cooler" (44.6% and 50.7%, respectively). Very few

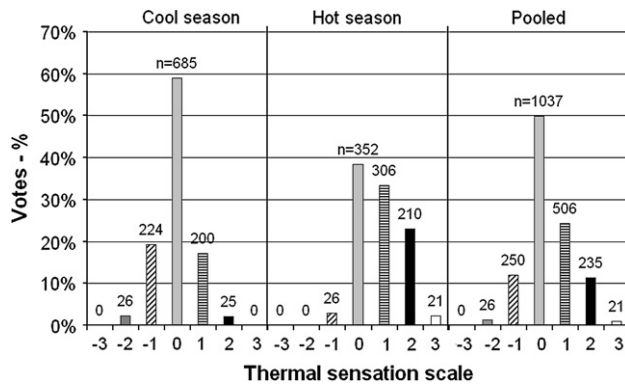


Fig. 4. Frequency distribution of thermal sensation votes (numerical values of -3, -2, -1, 0, 1, 2, and 3 indicate cold, cool, slightly cool, neutral, slightly warm, warm, and hot thermal sensations, respectively) for cool/hot season and also pooled.

subjects preferred to feel “warmer” (4.7%). Table 3 cross-tabulates percentages thermal preferences with air movement preferences. Almost half of the subjects preferred “more air movement” (49.2%) than they were experiencing at the time of their questionnaire. The remaining half of the sample was split into two different groups, with the majority preferring “no change” (44.5%) and only 6.1% requesting “less air movement”.

When crossed with thermal preference, subjects’ air movement preference for “more” were concentrated into “cooler” and “no change” thermal preferences. Those subjects indicating a preference for “less air movement” were concentrated in the “warmer” thermal preference group. Similar results were found for Zhang et al. [18] for office workers’ preferences for air movement from a database of indoor environmental quality surveys performed in over 200 buildings. According to their results, dissatisfaction with the amount of air motion is very common, with “too little air movement” cited far more commonly than “too much air movement”.

Our questionnaire requested subjects to assess the air movement within their work environment both in terms of *preference and acceptability*. Fig. 8 summarizes the overall air movement preferences binned according to air velocity values recorded at the time of the questionnaire. The percentage requesting “no change”

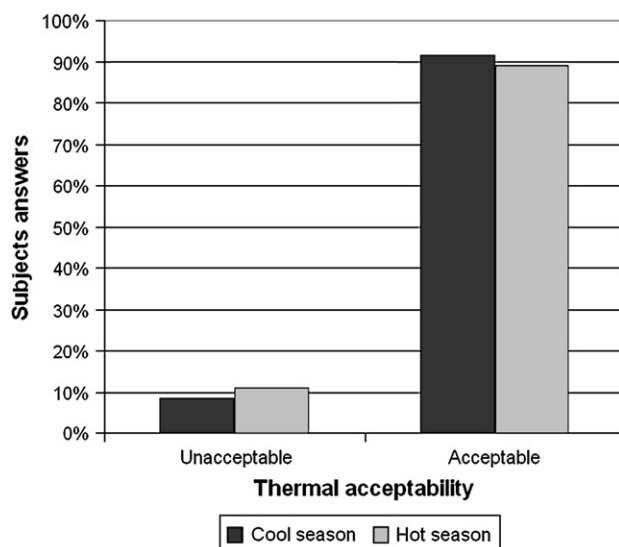


Fig. 5. Overall thermal acceptability for both seasons.

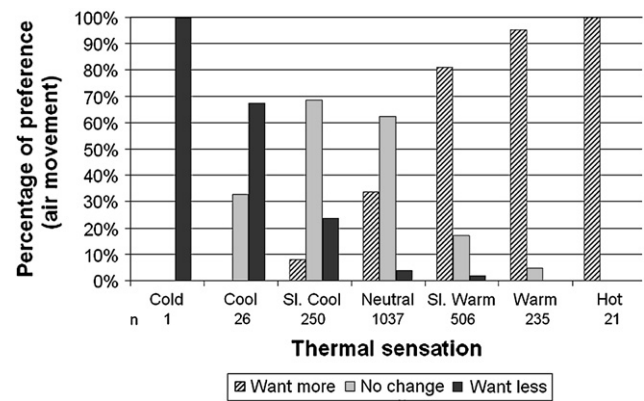


Fig. 6. Overall thermal sensation and air movement preference.

in air speed remained around 45% of subjects who were exposed to air speeds in the range 0.1–0.5 m/s, but then the percentage voting for “no change” in air speed increased at an almost linear rate as measured air speeds increased from 0.5 to 0.9 m/s. The percentage of subjects requesting “less air movement” remained below 10% across the entire range of measured air speeds. For those subjects asking for “more air movement”, the percentages demonstrate an opposite pattern in Fig. 8 to the “no change” votes described earlier (i.e. static rate of “want more” votes between 0.1 and 0.5 m/s air speeds, but then a steady decrease in the percentage of such requests as measured air speeds increased from 0.5 to 0.9 m/s).

Fig. 9 sorts the samples into those who found the air movement at the time of their questionnaire to be “acceptable” (Fig. 9a) and those who assessed it as “unacceptable” (Fig. 9b). In Fig. 9a (air movement acceptable) the percentage preferring “no change” in air movement increased as air velocity increased. On the other hand, the percentage of subjects asking for “more air movement” decreased with increasing air velocity. The number of subjects preferring “less air movement” remained below 10% across the entire velocity range. Based on this and in combination to the operative ranges, it was possible to identify the demand to higher air velocity values, even up to 0.8 m/s which is indicated as the maximum limit in ASHRAE 55 [12].

Fig. 9b summarizes the subjects who indicated that the air movement at the time of questionnaire was unacceptable combined with their air movement preference, binned according to measured air velocity values. For this group, the subjects expressed necessity for “more air movement” in a majority (maximum of 90% for 0.5 m/s and minimum of 45% for 0.9 m/s). The number of subjects requesting “less air movement” was in the minority, with a maximum percentage of 10% occurring at 0.7 and also 0.9 m/s.

Subjects were asked to assess air velocity acceptability and also to give their reasons (*too low, enough, or too high*). For both extremes (too low and too high air velocity), the subjects indicated

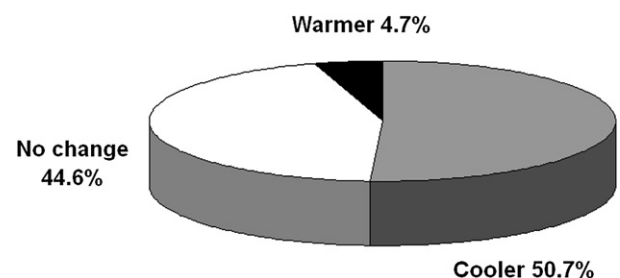


Fig. 7. Overall thermal preference.

Table 3
Cross-tabulated percentages for thermal and air movement preferences.

Thermal preference	Air movement preference					
	More		No change		Less	
	%	n	%	n	%	n
Warmer	6.0	8	44.0	59	50.0	67
No change	21.2	220	74.5	773	4.3	45
Cooler	88.3	797	10.1	91	1.7	15
Total	49.4	1025	44.5	923	6.1	127

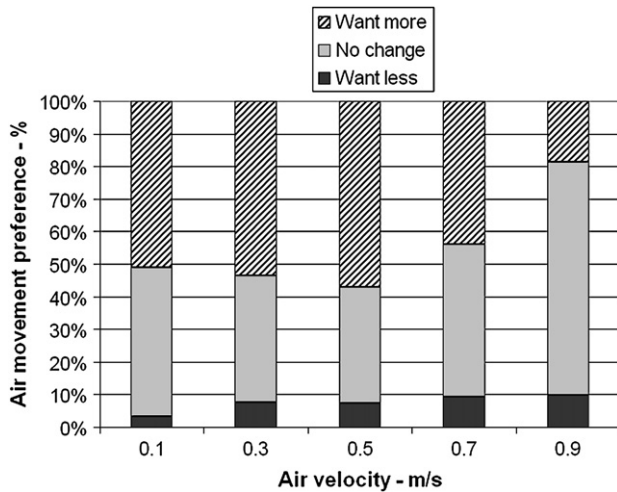


Fig. 8. Overall air movement preference and air velocity range.

their acceptability as described in Table 2 previously (see Section 2). The overall air velocity acceptability votes binned according to air velocity at the same time of questionnaire is indicated in Fig. 10. It is possible to identify a majority of the sample concentrated into the three *acceptable* categories (−1, 0 and 1).

For the “acceptable but too low air velocity” (−1) answers in Fig. 10, the values were approximately 50% at an air velocity of 0.1 m/s decreasing for 22% at 0.3 m/s, 15% at 0.5 m/s and less than 8% up to 0.7 m/s. As for the “acceptable but too high air velocity” answers, 35% of the sample was concentrated in the air velocities between 0.8 and 1.00 m/s. Of the two *unacceptable* categories, most fell into the “too low air velocity” rather than “too high air velocity”. Over 40% of the subjects exposed to air velocity <0.2 m/s assessed it as “unacceptable because of too low air velocity”. For air velocity in

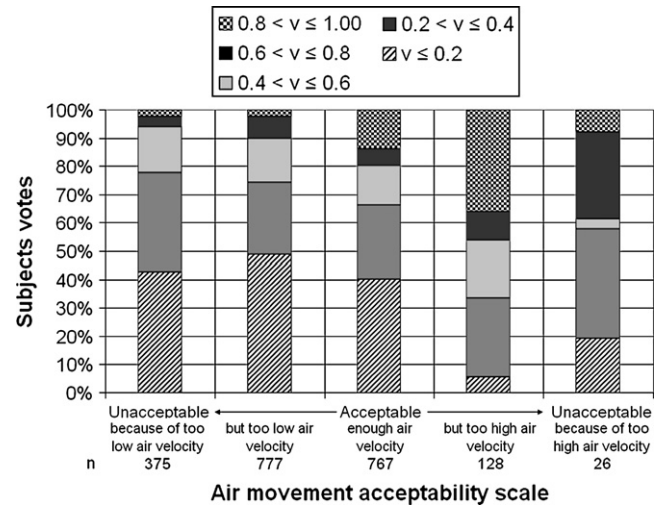


Fig. 10. Air velocity acceptability assessments within different prevailing air velocities.

the range $0.2 \leq v \leq 0.4$ m/s, this percentage decreased to 33% and decreased further to less than 15% for the air velocities up to 0.5 m/s. On the other hand, approximately 20% of those subjects voting unacceptable “because of too high air velocity”, were concentrated at air velocities less than 0.2 m/s, and another 35% were registered at velocities between 0.2 and 0.4 m/s.

As indicated earlier, the number of subjects assessing the air velocity at the time of questionnaire as being “too low” was overwhelmingly higher than those voting “too high”. These sub-samples were binned by operative temperature within each of the five air velocity ranges. For each degree of operative temperature (varying from 24 to 30 °C) the subjects’ air movement acceptability votes were binned into the five air velocities (from 0.1 to 0.9 m/s). Based on these cross-tabulations it was possible to identify minimal air velocity values necessary for air movement acceptability percentages of 80% (V_{80}) and also 90% (V_{90}). Air movement acceptability votes have been binned into 1 °C of operative temperature and 0.1 m/s air velocity intervals and the resulting percentages within each bin have been subjected to probit analyses. For this analysis only the three acceptable votes were used *acceptable but too low air velocity*, *enough air velocity* and *acceptable but too high air velocity*. The resulting Probit models are presented as curves in Fig. 11. The 25 °C Probit model of air movement acceptability has been omitted because the skewed distribution of the votes led to an insignificant Probit model.

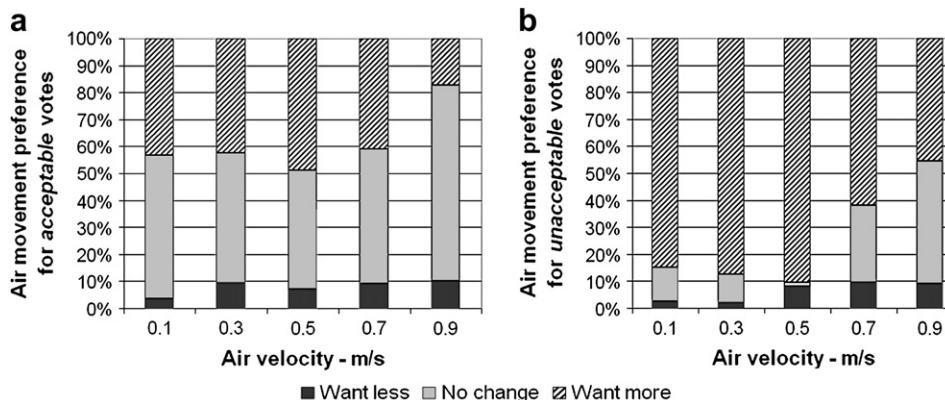


Fig. 9. Air movement preferences of those subjects for whom the air movement was (a) *acceptable* or (b) *unacceptable*.

According to ASHRAE 55 [12], “the required air speed may not be higher than 0.8 m/s” (lightly clothed person – clothing insulation between 0.5 clo and 0.7 clo who is engaged in near sedentary physical activity – metabolic rates between 1.0 met and 1.3 met). Fig. 11 and Table 4 set *minimal* air velocity values indicated as acceptable for 80% (V_{80}) of the subjects were near this maximum limit and above it for 90% of acceptability (V_{90}) for operative temperatures above 29 °C.

For air movement acceptability of 80% (V_{80}), the minimal air velocity required was 0.4 m/s at an operative temperature of 26 °C and for operative temperatures of 27 and 28 °C air velocity values required for 80% acceptability were 0.5 m/s. For operative temperatures of 29 °C and 30 °C air velocity values were slightly higher (0.6 and 0.7 m/s, respectively).

Considering an air movement acceptability of 90%, there is an increase in required air velocity values. For an operative temperature of 26 °C the air velocity required for 90% of acceptability (V_{90}) was 0.5 m/s. Minimal air velocity values required were equal to 0.6 m/s when operative temperature were 28 °C, 0.7 m/s for 29 °C and 0.9 m/s for 30 °C. For 30 °C of operative temperature, the maximum acceptability was 85% even when air velocities were increased for more than 0.9 m/s.

4. Discussion

In relation to thermal preference it is clear to identify that a majority voted for the maintenance of “no change” and “cooler”. When cross-tabulated percentages with air movement preferences, subjects voting for “more air movement” were concentrated in the “cooler” and “no change” categories. For those subjects indicating an air movement preference for “less air movement” there was a concentration of “want warmer” preference votes.

In relation to air movement preference, most subjects requested “more air movement” even in air speeds above 0.50 m/s. On the other hand, subjects who requested “less air movement” were few in number. These two generalizations combined indicate clearly that these Brazilian subjects prefer higher air speed values in order to improve their thermal comfort condition.

In addition, this study demonstrates a tolerance for air speeds up to 0.7 m/s. Subjects’ responses suggested that air movement and also air velocity were acceptable for the most part. Nevertheless we found a few cases in which the air movement was unacceptable and these were generally those subjects who indicated “cool” or “slightly cool” as their thermal sensation. Draft due to elevated air velocity values was much less than the opposite complaint of “too low” air velocity values. In summary, the main complaint was due

Table 4

Minimum air velocities required to achieve 80% and 90% acceptability in relation to operative temperature.

Operative temperature (°C)	Air movement acceptability	
	V_{80} , m/s	V_{90} , m/s
26	0.4	0.5
27	0.5	0.6
28	0.5	0.6
29	0.6	0.7
30	0.7	0.9
31	0.7	–

to “too low air velocity” and the percentages were overwhelmingly higher than those subjects’ classifying air movement as “too high”. Draft risk is definitely not the main complaint for these samples in naturally ventilated buildings in for a hot and humid climate such as Maceio city, Brazil.

In an attempt to identify subjects’ minimal air velocity value requirements, two different percentages were defined as goals for air movement acceptability: 80% and 90% (V_{80} and V_{90} , respectively). The minimal air velocities were at least 0.4 m/s for an operative temperature of 26 °C and rising to 0.9 m/s at 30 °C. Subjects are not only preferring more air movement but also indicating minimal air velocity values close or greater than the 0.8 m/s ASHRAE limit. These findings suggest a strong demand for air movement inside naturally ventilated buildings but also a tolerance for higher air velocity values.

5. Conclusions

Our results lead to the conclusion that air movement can be quite acceptable at speeds well above the previous values suggested in the literature. For natural ventilation in hot and humid climates, higher air speeds may be desirable in order to improve subjects’ thermal comfort. This dispels the notion of draft in hot and humid climates and it is consistent with the broader theory of *alliesthesia* and the physiological role of pleasure due to air movement increment. By linking the physiological concept of *alliesthesia* with knowledge about cutaneous thermoreceptors it is possible to understand the simple *pleasure* that we derive from effective natural ventilation, particularly in warm climates [6]. These findings also corroborate to previous studies addressed to the pleasantness associated with transient conditions [19,20].

Subjects preferring “more air movement” were significantly more numerous than those demanding “less air movement”. The majority of subjects considered air movement “acceptable”. For the minority percentage classifying air movement as “unacceptable”, their main reason was “too low air movement”. Based on this strong demand for more air movement, subjects’ acceptability was investigated based on two goals for movement acceptability (80 and 90%). Minimal air velocity values obtained based on these goals were close to or above 0.8 m/s which is considered as the maximum air velocity for ASHRAE 55 [12].

These results suggest that subjects’ acceptance of higher air velocities increased to compensate for elevated temperature and humidity. In summary, air movement can be quite acceptable at speeds well excess of the previous values suggested in the literature and standards. Focusing in future Brazilian standards, these results suggested the necessity of more experiments related to minimum air velocity requirements for naturally ventilated environments.

In addition, it is important that the occupants should be able to control the airflow inside the buildings according to their personal preferences. Future experiments should be carried out in order to identify air movement acceptability inside indoor environments, which differ from the ones investigated into this research, such as office and residential buildings.

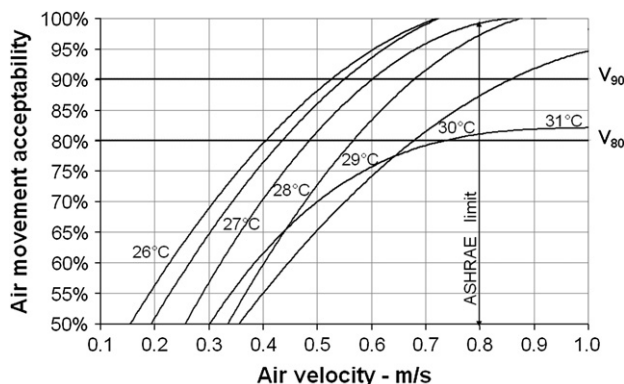


Fig. 11. Air movement acceptability and operative temperature binned by air velocity values resulted from the probit regression analysis (95% confidence levels).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.buildenv.2009.06.005](https://doi.org/10.1016/j.buildenv.2009.06.005).

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